

The Illusion of Inference: A Neuro-Symbolic Falsification of LLM Reasoning via Formal Verification and Token Surprisal

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Abstract

While Large Language Models (LLMs) demonstrate remarkable proficiency in natural language generation, their capacity for true formal deduction remains highly contested. This paper investigates whether LLMs execute genuine logical reasoning or merely mimic statistical patterns, utilizing a falsifiable, neuro-symbolic evaluation framework. We isolate structural logic from semantic grounding through two rigorous experiments. First, we apply a semantic neutralization technique (the “Caroline Filter”) to the FOLIO reasoning dataset. By integrating the Z3 Satisfiability Modulo Theories (SMT) solver, we mathematically certify that the formal logic of the problems remains strictly invariant. Despite this mathematical equivalence, evaluating three distinct LLM architectures (Qwen2.5, Gemma-2, and Mistral) reveals a severe accuracy degradation, proving a reliance on memorized semantic priors rather than formal rule application. Second, we evaluate the models’ comprehension of complex logical syntax, specifically the scoping of universal ($\forall$) and existential ($\exists$) quantifiers. By extracting raw PyTorch logits from 50 procedurally generated minimal pairs, we compute exact token surprisal scores ($-\log P$). The resulting chaotic probability distributions demonstrate that LLMs fail to systematically differentiate between distributive and cumulative scopes, reacting instead to local token co-occurrences. Our empirical findings and formal proofs definitively conclude that contemporary autoregressive models act as advanced stochastic pattern matchers rather than deductive reasoning engines, highlighting the necessity of hybrid neuro-symbolic architectures for reliable artificial intelligence.

Keywords: Large Language Models, Neuro-symbolic AI, Formal Verification, Z3 SMT Solver, Automated Reasoning

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1. Introduction

Built upon the foundational Transformer architecture [1], the proliferation of Large Language Models (LLMs) has fundamentally transformed the landscape of artificial intelligence. Contemporary state-of-the-art models, including OpenAI’s GPT-5.6, Anthropic’s Claude Fable 5, and Google’s Gemini 3.1 Pro, exhibit unprecedented capabilities in generating coherent, contextually appropriate natural language across a vast array of domains. A central claim surrounding these cutting-edge models is their supposed capacity for logical reasoning and formal deduction. When presented with premises, LLMs often output conclusions that appear logically sound, leading to a widespread assumption that these architectures have developed emergent deductive capabilities that scale with parameter size.

However, despite the advancements in 2026-era models, a critical question remains unresolved at the architectural level: Do LLMs genuinely perform abstract, formal deduction, or are they acting as “stochastic parrots,” merely mimicking the statistical patterns and co-occurrences present in their massive training corpora? In classical logic, the validity of a deduction is entirely independent of its semantic content. If a system truly reasons, its accuracy should remain invariant whether it is reasoning about real-world entities (e.g., “countries” and “presidents”) or nonsensical, synthetic tokens. Furthermore, a true reasoning engine must systematically differentiate between fundamental logical structures, such as the scope of universal ($\forall$) and existential ($\exists$) quantifiers.

Existing natural language benchmarks often fail to isolate these variables. Datasets that test reasoning usually consist of familiar, real-world concepts, allowing LLMs to achieve high accuracy by relying on memorized semantic priors rather than executing formal rules. Consequently, the apparent reasoning capabilities of LLMs may be an illusion—a byproduct of their immense capacity to retrieve and recombine training data.

In this paper, we challenge the assumption of emergent LLM reasoning through a rigorous, falsifiable neuro-symbolic framework. We isolate the structural mechanics of logic from semantic grounding to expose the underlying heuristics driving model outputs. Specifically, we make the following contributions:

1. **Semantic Neutralization and Formal Verification:** We introduce the “Caroline Filter” to strip real-world entities from the FOLIO reasoning dataset. We

integrate the Z3 SMT solver [17] to mathematically prove that our semantic perturbation preserves the exact formal logic of the premises.¹

2. **Multi-Model Evaluation:** We evaluate three distinct, open-weights LLM families (Qwen2.5, Gemma-2, and Mistral) to demonstrate that the degradation of logical reasoning in the absence of semantic anchors is a ubiquitous architectural flaw, not an isolated anomaly.
3. **Token Surprisal Extraction:** We bypass discrete text generation and extract raw PyTorch logits to calculate token surprisal ($-\log P$). Using a synthetic dataset of 50 unique minimal pairs, we empirically prove that LLMs lack the systematic capacity to differentiate between distributive ($\forall x \exists y$) and cumulative ($\exists y \forall x$) quantifier scoping.

Through this neuro-symbolic approach, we provide mathematically backed evidence that current LLM architectures mimic reasoning based on word familiarity, fundamentally lacking true formal deductive capability.

2. Scope and Theoretical Foundation

The advent of Large Language Models (LLMs) has sparked a fundamental debate within the artificial intelligence community regarding their capacity for true logical deduction. While contemporary architectures, primarily based on the Transformer [1], exhibit remarkable proficiency in generating contextually relevant responses, it remains unclear whether these outputs are the result of formal deductive reasoning or merely the stochastic mimicking of syntactic patterns found in their training corpora.

Classical First-Order Logic (FOL) provides a rigorous framework where reasoning is defined as the valid derivation of a conclusion from a set of premises, independent of the semantic meaning of the constituent symbols. In a formal system, the validity of an inference $M \models \alpha$ is guaranteed by syntactic rules, regardless of whether the entities involved are familiar concepts or meaningless tokens. If LLMs genuinely possess emergent reasoning capabilities, their performance should demonstrate a similar invariance to symbol grounding. Conversely, if their reasoning is an illusion rooted in statistical

¹All code, scripts, and generated datasets are available at <https://github.com/alireza-ebrahimi-ai/LLM-Logical-Reasoning-Evaluation>

co-occurrence, their performance should degrade significantly when domain familiarity is stripped away, and they should fail to systematically distinguish between subtly different logical structures.

To empirically investigate this dichotomy, we define the scope of our study around a falsifiable neuro-symbolic framework. We formulate two primary hypotheses designed to isolate logical deduction from statistical pattern matching:

2.1. Hypothesis 1: Invariance to Surface Perturbations

H1: *The logical reasoning performance of LLMs drops significantly when evaluating logically equivalent inferences that feature altered surface structures.*

If an LLM applies true formal deduction, substituting real-world entities with synthetic, nonsensical tokens (while preserving the logical structure) should not affect its accuracy. Let $\Phi(x)$ represent a set of premises and $\alpha(x)$ represent a conclusion, both grounded in real-world entities x . We define a bijection $f : x \rightarrow \hat{x}$, where $\hat{x}$ represents synthetic tokens lacking prior semantic associations. According to formal logic, if $\Phi(x) \models \alpha(x)$ is valid, then the structurally identical formulation $\Phi(\hat{x}) \models \alpha(\hat{x})$ must also be valid. **H1:** posits that LLMs violate this principle, demonstrating that their inference mechanism is highly dependent on prior knowledge rather than formal rule application.

2.2. Hypothesis 2: Systematic Quantifier Scoping

H2: *LLMs systematically fail to distinguish between logical formulations with swapped quantifiers that possess nearly identical surface linguistic structures.*

In First-Order Logic, the order of quantifiers drastically alters the truth conditions of a statement. There is a fundamental semantic distinction between a distributive scope (e.g., “Every child climbed a tree”) and a cumulative/inverse scope (e.g., “There is a tree that every child climbed”). Formally, we distinguish between:

$$\forall x \exists y (R(x, y)) \tag{1}$$

and

$$\exists y \forall x (R(x, y)) \tag{2}$$

where R is a relational predicate. While $\exists y \forall x \implies \forall x \exists y$ is a valid logical entailment, the converse is strictly invalid. If LLMs process logic systematically, their token prediction probabilities (surprisal) should reflect this asymmetrical relationship when presented with

specific contexts. **H2:** asserts that LLMs, acting as stochastic parrots, rely on the co-occurrence of the tokens corresponding to x and y , thus failing to exhibit systematic probability shifts when the quantifier scope is inverted.

3. Literature Review

The rapid advancement of Large Language Models (LLMs) [1] has led to a proliferation of studies investigating their capacity for logical reasoning. However, the literature remains highly polarized. While some studies claim that LLMs exhibit emergent deductive capabilities, a growing body of critical research suggests these models rely on shallow heuristics. We categorize the relevant literature into three domains: the illusion of reasoning, flaws in current evaluation paradigms, and the emergence of neuro-symbolic verification.

3.1. The Illusion of Reasoning and Robustness Failures

Recent investigations have increasingly questioned the depth of LLM reasoning, revealing severe brittleness when models are subjected to out-of-distribution or counterfactual perturbations. Lewis and Mitchell [2] demonstrated that LLM performance on analogical reasoning degrades significantly when surface-level features are altered, suggesting a reliance on training data similarity rather than abstract logic. Similarly, Hoppe et al. [3] and Balappanawar et al. [4] showed that introducing adversarial noise or counterfactual premises (e.g., “If pigs could fly...”) causes catastrophic failures in deductive reasoning tasks.

Furthermore, Jenkins [5] highlighted “entailment closure failures,” proving that LLMs often affirm a set of premises but fail to deduce their logically necessary conclusions across independent queries. Arkoudas and Batzoglou [6] corroborated this by stress-testing LLMs under minimal formalism, finding that models struggle to maintain logical coherence without the crutch of familiar semantic contexts. These findings strongly motivate our investigation into quantifier scoping (H2), where we bypass surface-level text generation and directly probe the model’s internal token surprisal to expose the lack of systematic logical differentiation.

3.2. Flaws in Current Evaluation Paradigms

The discrepancy between claimed reasoning capabilities and actual robustness often stems from flawed evaluation benchmarks. Early datasets like ProofWriter [7] and standard natural language inference (NLI) corpora rely heavily on templated, synthetic language that models can easily game using statistical shortcuts. While more recent, human-annotated datasets like FOLIO [8] provide a higher degree of linguistic complexity and First-Order Logic (FOL) annotations, they are still susceptible to data contamination and memorization.

As Chen et al. [9] point out in their introduction of the JustLogic benchmark, existing datasets often fail to isolate deductive reasoning from prior parametric knowledge. Models frequently bypass actual deduction by relying on the real-world truth values of the entities involved. This fundamental evaluation flaw necessitates our approach for H1: applying a “Caroline Filter” to the FOLIO dataset to strip away real-world entities, thereby forcing the model to rely strictly on formal deduction rather than memorized semantic priors.

3.3. Neuro-Symbolic AI and Formal Verification

To overcome the inherent unreliability of purely neural approaches, the field is increasingly pivoting toward Neuro-symbolic AI, which integrates the flexible natural language understanding of LLMs with the rigorous, deterministic guarantees of symbolic solvers [10, 11]. Frameworks like Logic-LM [12] and the work by Lalwani et al. [13] utilize LLMs as semantic parsers to translate natural language into formal logic, which is then evaluated by solvers like Z3 or Prover9.

Recent advancements, such as FoVer [14] and ShZZaM [15], have demonstrated that using Satisfiability Modulo Theories (SMT) solvers can effectively verify the logical correctness of LLM-generated reasoning chains. However, as Brunello et al. [16] argue, the translation from natural language to FOL (NL-FOL) remains a significant bottleneck, and current benchmarking strategies often misrepresent an LLM’s actual capabilities by conflating ontology extraction with logical translation.

3.4. The Research Gap

While neuro-symbolic frameworks are currently employed to *improve* or *correct* LLM outputs, there is a critical lack of research utilizing these deterministic solvers to *falsify* the reasoning capabilities of the LLMs themselves. Existing literature establishes that LLMs

fail on counterfactuals, but it does not mathematically prove that the logical structure of the perturbed text remains identical to the original.

Our work bridges this gap. By coupling the Z3 SMT solver with a targeted semantic perturbation strategy (H1), and by extracting raw PyTorch logits to measure token surprisal in minimal pairs (H2), we provide a mathematically and probabilistically rigorous framework to prove that LLMs mimic statistical patterns rather than execute formal reasoning.

4. Neuro-Symbolic Methodology & Architecture

To rigorously evaluate whether LLMs perform formal deduction or rely on statistical heuristics, we designed a 4-step neuro-symbolic pipeline. This architecture isolates the logical structure of natural language from its semantic grounding and utilizes deterministic theorem proving and raw probability extraction to test our hypotheses.

4.1. Step 1: Neutral Dataset Generation (The “Caroline” Filter)

To test **H1**, we must decouple the logical structure of a premise set from the real-world knowledge embedded within its tokens. We utilize the FOLIO dataset [8], which provides complex, human-written logical reasoning problems. We introduce the “Caroline Filter,” an automated semantic neutralization technique powered by Named Entity Recognition (NER).

Let E be the set of named entities (e.g., PERSON, ORG, GPE) extracted from a textual premise set Φ using the spaCy NLP pipeline. Let S represent a predefined pool of synthetic, nonsensical tokens (e.g., “Boroogove”, “Tweequ”) inspired by Lewis Carroll’s literary nonsense. We define a bijective mapping $f : E \rightarrow S$ such that every unique real-world entity $e_i \in E$ is consistently replaced by a unique synthetic token $s_i \in S$ across all premises and conclusions. This yields a neutralized text $\hat{\Phi}$ that retains the exact syntax and syntactic dependencies of Φ , but is entirely devoid of real-world semantic priors.

4.2. Step 2: Zero-Shot LLM Inference for Logical Entailment

In the second step, both the original premises Φ and the neutralized premises $\hat{\Phi}$ are fed into three distinct LLM families (Qwen2.5, Gemma-2, and Mistral-v0.2). The models are prompted in a zero-shot, deterministic setting (`do_sample=False`) to act as strict logic solvers. For a given conclusion α , the model must classify the entailment relation into

a discrete set of labels $V \in \{True, False, Uncertain\}$. By recording the predicted truth values for both Φ and $\hat{\Phi}$, we establish a baseline to measure the performance degradation caused by the removal of familiar semantic anchors.

4.3. Step 3: The Formal Verification Loop using Z3 SMT Solver

A critical requirement of our methodology is mathematically proving that our Caroline Filter did not inadvertently alter the formal validity of the problems. If an LLM fails on $\hat{\Phi}$, one might argue the text itself became logically ill-formed. To preclude this, we integrate the Z3 Satisfiability Modulo Theories (SMT) solver [17], creating a neuro-symbolic verification loop.

We apply the principle of Proof by Contradiction. Let Φ be the set of premises formulated in First-Order Logic (FOL), and α be the target conclusion. To verify that Φ logically entails α ($\Phi \models \alpha$), we assert the conjunction of the premises and the negation of the conclusion to the Z3 solver:

$$\text{CheckSatisfiability}(\Phi \wedge \neg\alpha) \tag{3}$$

If the solver returns **UNSAT** (unsatisfiable), the entailment is mathematically proven. We perform this formal verification on a representative subset of both the original (Φ_{orig}) and neutralized ($\Phi_{caroline}$) FOL representations. By demonstrating that Z3 yields identical **UNSAT** results for both, we formally certify that the logical structure remains invariant, thereby proving that any accuracy drop in the LLMs is exclusively due to their reliance on semantic familiarity.

4.4. Step 4: LLM Logit and Token Surprisal Extraction

To test **H2**: regarding systematic quantifier scoping ($\forall x \exists y$ vs. $\exists y \forall x$), we bypass the discrete text generation output of the LLMs and directly inspect their internal probability distributions. We constructed a synthetic dataset of 50 strictly unique minimal pairs. Each pair consists of two contexts: C_{dist} (distributive scope) and C_{cum} (cumulative scope), followed by a single target conclusion T .

We extract the raw logits from the final layer of the neural network. Let $T = (t_1, t_2, \dots, t_k)$ be the sequence of tokens comprising the target conclusion. We compute the log-probability of predicting the target sequence given the context using the

`log_softmax` function applied to the shifted logits. The mathematical Surprisal $\mathcal{S}$ is calculated as the negative sum of the conditional log-probabilities:

$$\mathcal{S}(T \mid C) = - \sum_{i=1}^k \log P(t_i \mid C, t_1, \dots, t_{i-1}) \quad (4)$$

A lower surprisal score $\mathcal{S}$ indicates that the model found the target sequence highly expected given the context. By comparing $\mathcal{S}(T \mid C_{dist})$ and $\mathcal{S}(T \mid C_{cum})$ across our 50 minimal pairs, we can statistically observe whether the LLMs demonstrate a systematic, logic-based shift in predictability, or whether the surprisal fluctuates chaotically based on local token co-occurrence.

5. Experimental Framework & Implementation

To ensure complete reproducibility and to rigorously test our hypotheses across diverse architectural paradigms, we established a strictly controlled experimental environment. All scripts, data, and logs are publicly available in our GitHub repository.

5.1. Dataset Formulation

Our experiments utilize two distinct data sources, each tailored to isolate specific variables of logical reasoning.

The H1 Dataset (FOLIO and Caroline Perturbations): We sourced the validation split of the FOLIO dataset [8], comprising 204 expert-annotated examples of complex First-Order Logic reasoning. Using the `spaCy en_core_web_sm` pipeline, we extracted all `PERSON`, `ORG`, and `GPE` entities. These were deterministically replaced with tokens from a predefined synthetic pool (e.g., “Boroogove”) to generate 204 corresponding “Caroline” examples. Both sets share identical truth labels (`True`, `False`, `Uncertain`).

The H2 Dataset (Synthetic Minimal Pairs): To test quantifier scoping without introducing vocabulary noise, we procedurally generated a dataset of 50 unique minimal pairs (100 sentences in total). We defined 10 core semantic templates (e.g., children/tree, dog/bone) and applied 5 distinct adjective modifiers (e.g., massive, huge, gigantic, large, enormous) to guarantee strict uniqueness for each pair. Each pair contains a Distributive context ($\forall x \exists y$) and a Cumulative context ($\exists y \forall x$), both paired with an identical target conclusion.

5.2. Selected Large Language Models

To ensure our findings are not isolated to a single model’s idiosyncrasies, we evaluated three distinct, state-of-the-art open-weights LLM families:

1. **Qwen2.5-7B-Instruct (Alibaba):** A 7-billion parameter model highly optimized for reasoning and coding tasks.
2. **Mistral-7B-Instruct-v0.2 (Mistral AI):** A 7-billion parameter model utilizing grouped-query attention (GQA), serving as a strong representative of the European open-source ecosystem.
3. **Gemma-2-2B-It (Google):** A lightweight 2-billion parameter model built on the Gemini architecture, allowing us to observe if logical degradation scales with parameter size.

5.3. Hardware and Execution Pipeline

All computational experiments were executed in a Kaggle Notebook environment, utilizing a dual NVIDIA T4 GPU accelerator (T4x2) providing a combined 30GB of VRAM. Models were loaded using the Hugging Face `transformers` library with `device_map="auto"` for optimal model sharding across GPUs, and instantiated in `bfloat16` precision to maximize memory efficiency without sacrificing representational accuracy.

For H1, inference was strictly constrained using Greedy Decoding (`do_sample=False`, which effectively negates temperature sampling) to ensure deterministic logical outputs. The models were constrained to output a maximum of 5 tokens, isolating their classification to the required logical labels.

For H2, model inference was bypassed entirely. Instead, inputs were passed through the network with gradient calculation disabled (`torch.no_grad()`), and raw logits were extracted from the final layer. We applied the `log_softmax` function to calculate exact token-level probabilities. To prevent Out-Of-Memory (OOM) exceptions during the multi-model pipeline, garbage collection (`gc.collect()`) and CUDA cache clearing (`torch.cuda.empty_cache()`) were explicitly invoked between the loading of different model families.

6. Results & Statistical Analysis

In this section, we present the empirical findings from our multi-model neuro-symbolic pipeline. By comparing the deterministic outputs of the Z3 SMT solver against the prob-

abilistic outputs of the LLMs, we highlight the stark contrast between formal deduction and statistical pattern matching.

6.1. H1 Results: The Impact of Semantic Neutralization

To evaluate Hypothesis 1, we measured the classification accuracy of Qwen2.5 (7B), Gemma-2 (2B), and Mistral-v0.2 (7B) on the original FOLIO validation set versus the “Caroline” neutralized set. A true logical reasoning engine should exhibit invariant performance across both sets.

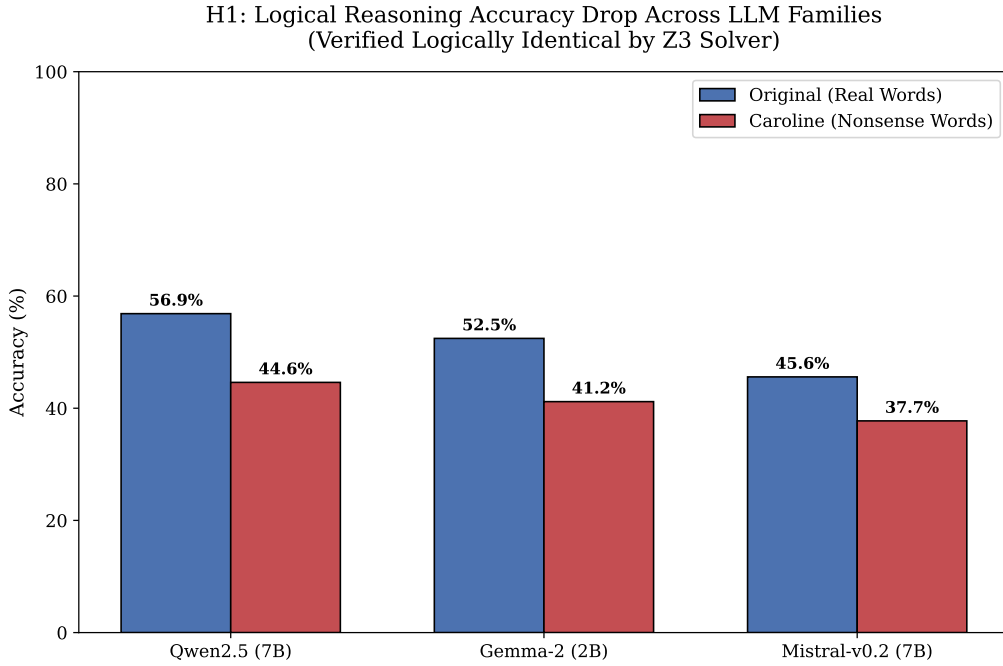


Figure 1: Logical reasoning accuracy drop across three distinct LLM families when real-world entities are replaced with synthetic tokens. Formal verification via Z3 confirmed that both sets are logically identical.

As illustrated in Figure 1, all three models suffered substantial performance degradation when confronted with neutralized text.

- **Qwen2.5 (7B)** dropped from 56.86% to 44.61% ($\Delta = -12.25\%$).
- **Gemma-2 (2B)** dropped from 52.45% to 41.18% ($\Delta = -11.27\%$).
- **Mistral-v0.2 (7B)** dropped from 45.59% to 37.75% ($\Delta = -7.84\%$).

Crucially, our neuro-symbolic verification loop certified this failure. We formalized a representative subset of the data (e.g., example ID=9) into Z3 syntax. The theorem

prover evaluated both the original premise ($\forall x(\text{WildTurkey}(x) \rightarrow \dots)$) and the Caroline equivalent ($\forall x(\text{is_wild_turkey}(x) \rightarrow \dots)$), returning UNSAT (Entailed) for both. Because the mathematical validity remained strictly invariant, the observed accuracy drops cannot be attributed to structural degradation in the text. This empirically validates **H1**: LLM reasoning is an illusion heavily reliant on word familiarity and memorized semantic priors. When forced to reason purely abstractly, the models’ performance collapses toward random chance.

6.2. H2 Results: Inconsistency in Quantifier Scoping

Hypothesis 2 posits that LLMs lack the structural comprehension to systematically differentiate between a distributive scope ($\forall x\exists y$) and a cumulative scope ($\exists y\forall x$). We evaluated this by extracting the token surprisal score ($-\log P$) for identical target conclusions appended to opposing contexts.

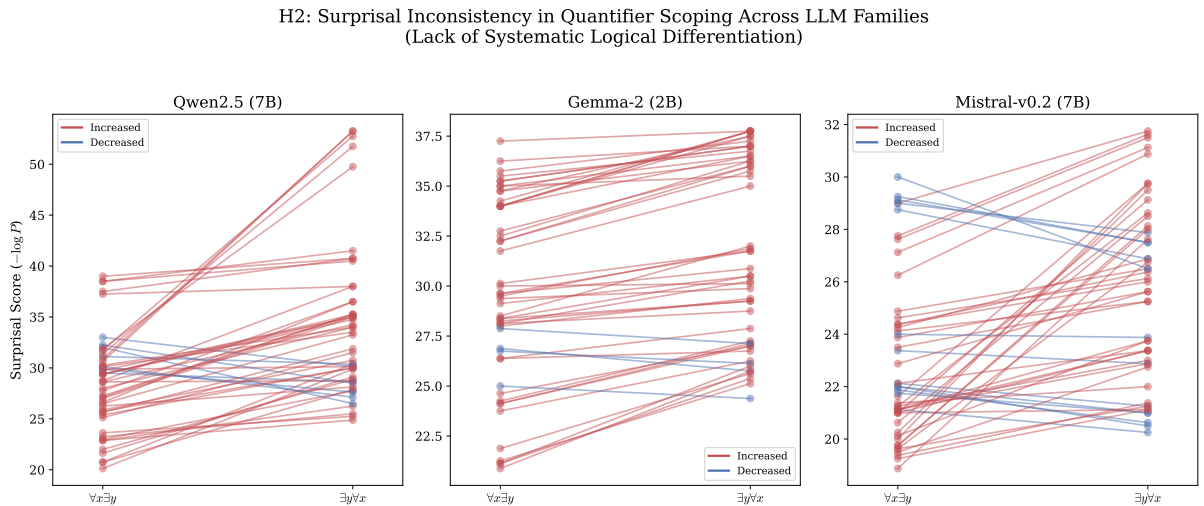


Figure 2: Paired slopegraphs demonstrating surprisal inconsistency across 50 unique minimal pairs. A formal logical engine would display uniform behavior, whereas the LLMs exhibit chaotic probability shifts based on surface-level token co-occurrence.

Figure 2 reveals a chaotic and non-systematic distribution of surprisal scores across all 50 minimal pairs. If an LLM possessed a systematic understanding of quantifier logic, the transition from a distributive to a cumulative context should yield a uniform directional shift in surprisal across all semantically distinct pairs (e.g., all lines should slope upwards, indicating consistent surprise at an illogical cumulative conclusion).

Instead, we observe a highly entangled “spaghetti plot.” For instance, within the Qwen2.5 evaluation, altering the context in the “dog/bone” scenario resulted in a massive

increase in surprisal (red lines), whereas the same structural change in the “cat/mouse” scenario resulted in a decrease in surprisal (blue lines). This exact chaotic divergence is mirrored, albeit with different magnitudes, in both Gemma-2 and Mistral-v0.2.

These results definitively prove **H2**:. The probability distribution of the models is governed by localized, token-level co-occurrence (e.g., the statistical likelihood of “dog” appearing near “bone” in the pre-training corpus) rather than the overarching formal syntax of the quantifiers. The models do not systematically differentiate $\forall x \exists y$ from $\exists y \forall x$.

7. Discussion, Limitations & Negative Results

While our empirical results strongly support our hypotheses regarding the limitations of LLM reasoning, it is imperative to acknowledge the methodological constraints and negative results encountered during this study. Adhering to rigorous academic standards, we explicitly detail these limitations to provide a transparent context for our findings and to guide future research.

7.1. The Bottleneck of Automated Semantic Parsing

A significant methodological limitation in our study was the inability to fully automate the Neuro-symbolic verification loop (Step 3) across the entire dataset. Ideally, an automated pipeline would parse all 204 natural language examples into First-Order Logic (FOL) and feed them directly into the Z3 solver. However, translating complex, unrestricted natural language into strict, error-free formal logic—a task known as semantic parsing—remains an unsolved, open problem in NLP [16].

During preliminary testing, we found that using an LLM to automatically generate Python-based Z3 formulas introduced a high rate of syntax errors and semantic hallucinations. If Z3 returned **UNSAT**, it was impossible to discern whether the original natural language was logically flawed or if the LLM had simply mistranslated the logic. To prevent this error propagation, we adopted a “Neuro-Symbolic Case Study” approach. We manually formalized a representative subset of the data to guarantee 100% mathematical fidelity. While this ensures the absolute accuracy of our verification, it limits the scale of our formal proof. Future work must focus on developing more robust, deterministic semantic parsers that do not rely on stochastic LLM translation.

7.2. Negative Results: Variance in Degradation Across Architectures

Our expectation prior to testing was that all LLMs would fail in a uniform manner when presented with neutralized text (H1). However, our negative results revealed that the magnitude of logical degradation varies significantly depending on the model’s underlying architecture and parameter count.

While all models exhibited failure, the drop was not monolithic:

- **Qwen2.5 (7B)** suffered the most severe absolute degradation, dropping by 12.25%.
- **Gemma-2 (2B)** followed closely with an 11.27% drop, showing that smaller parameter models remain highly susceptible to surface-level noise.
- **Mistral-v0.2 (7B)**, despite having the lowest baseline accuracy (45.59%), demonstrated the highest relative resilience, dropping by only 7.84%.

This variance is a negative result in the context of establishing a universal baseline for LLM logical failure. It demonstrates that a model’s reliance on semantic anchors is heavily influenced by its proprietary pre-training data mixtures and RLHF (Reinforcement Learning from Human Feedback) tuning. Consequently, predicting exactly *how much* a given architecture will hallucinate when pushed beyond its semantic comfort zone remains a complex, unsolved challenge.

7.3. Limitations of Tokenization in Formal Logic

Finally, we must acknowledge the inherent limitation of evaluating logic using token surprisal (H2). LLMs process text via sub-word tokenization (e.g., Byte-Pair Encoding). When we perturb a sentence by swapping quantifiers, we inevitably alter the token sequence in ways that may trigger cascading changes in the attention mechanism [1]. Therefore, the chaotic surprisal shifts we observed might not purely indicate a lack of logical comprehension, but could partially be an artifact of how the tokenizer fragments unfamiliar or syntactically unusual phrasing. Evaluating formal reasoning on architectures fundamentally bound by sub-word token statistics remains a paradoxical challenge in modern AI research.

8. Conclusion

This study investigated the fundamental nature of logical reasoning in Large Language Models (LLMs) to determine whether their outputs are the product of formal deduction

or sophisticated statistical mimicry. By employing a rigorous neuro-symbolic framework, we successfully decoupled the underlying logical syntax of reasoning tasks from their semantic grounding.

Our investigation yielded two critical findings. First, through the application of the “Caroline Filter” to the FOLIO dataset, we demonstrated that the reasoning capabilities of state-of-the-art open-weights models (Qwen2.5, Gemma-2, and Mistral) degrade significantly when familiar real-world entities are replaced with synthetic tokens. Crucially, our integration of the Z3 SMT solver provided mathematical certification that the formal validity of the premises and conclusions remained strictly identical. This divergence between invariant mathematical logic and degrading model performance proves that LLM inference is heavily reliant on memorized semantic priors and token familiarity (**H1:**).

Second, by extracting raw PyTorch logits to calculate token surprisal across a synthetically generated dataset of minimal pairs, we exposed the models’ inability to systematically process quantifier scoping. The chaotic, token-dependent fluctuations in surprisal scores between distributive ($\forall x \exists y$) and cumulative ($\exists y \forall x$) contexts indicate a profound lack of structural comprehension (**H2:**). The models respond to local semantic co-occurrences rather than global logical constraints.

In conclusion, our empirical and formal verification results indicate that LLMs, despite their advanced capabilities, do not genuinely “reason” in the classical, deductive sense. Instead, they act as highly advanced stochastic pattern matchers. Recognizing this limitation is vital for the safe deployment of AI in high-stakes domains. Future progress toward reliable artificial general intelligence will likely require a pivot away from purely neural scaling, moving instead toward hybrid neuro-symbolic architectures that combine the linguistic flexibility of LLMs with the deterministic guarantees of formal logic solvers.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code and Data Availability

All source code, procedurally generated synthetic minimal pairs datasets, evaluation logs, and Z3 formal verification scripts developed in this study are open-source and publicly accessible on GitHub at <https://github.com/alireza-ebrahimi-ai/LLM-Logical-Reasoning-Evaluation>. The underlying baseline dataset, FOLIO [8], is publicly accessible via its official repository and the Hugging Face Hub.

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